Algebra II with Trigonometry Chapter 7.3

Olympiad Problems

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Problems

1. Given that $\sin\theta + \sin 3\theta + \sin 5\theta = 0$, find the exact value of $\cos 2\theta \cdot \cos 4\theta$.

2. If $\tan x = \sqrt{7} + 2\sqrt{3}$, compute $\sin 2x$ in simplest radical form.

3. Without using a calculator, evaluate $\frac{\sin 75^\circ + \sin 15^\circ}{\cos 75^\circ + \cos 15^\circ}$.

4. Suppose $\sin^4 x + \cos^4 x = \frac{5}{8}$. Find the value of $\sin^2 2x$.

5. Let $t = \tan^{-1}\frac{1}{3}$. Find the exact value of $\sin 2t + \cos 2t$.

6. If $0 < x < \frac{\pi}{2}$ and $\cos x = 1 - 2\sin^2\frac{x}{2}$, determine the value of $\sin x$.